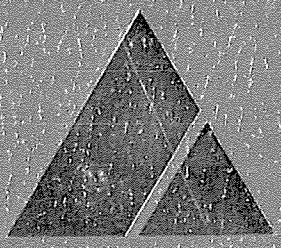
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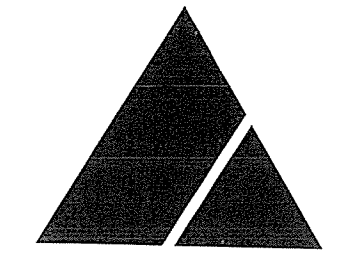


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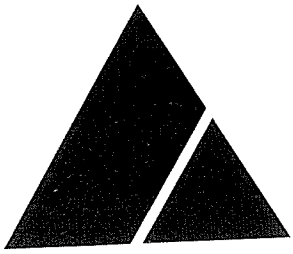


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ASPHALT HANDBOOK

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MANUAL SERIES NO. 4 (MS-4)

The Asphalt Institute can accept no responsibility for the inappropriate use of this manual. Engineering judgment and experience must be used to properly utilize the principles and guidelines contained in this manual, taking into account available equipment, local materials and conditions.

All reasonable care has been taken in the preparation of this manual; however, the Asphalt Institute can accept no responsibility for the consequence of any inaccuracies it may contain.

Printed in USA
Seventh Edition 2007
ISBN 978-1-934154-27-4
Library of Congress Control Number: 2007926839
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Asphalt Institute

Foreword

MS-4 is the Asphalt Institute's comprehensive manual on the use of asphalt. For 70 years, it has served the asphalt industry as the primary reference manual for contractors, engineers, consultants, specifiers, and user agencies.

With this printing, the Asphalt Institute is pleased to offer the seventh edition of this essential asphalt publication. This new, expanded edition of over 800 pages of text and illustrations showcases the advances in asphalt technology that have evolved since the 1989 edition of MS-4.

Readers of this edition will learn how to use the latest technologies to address a variety of modern pavement challenges, including durability, heavy loads, high traffic volumes, noise, smoothness, and safety to meet the demands of today's driving public.

This seventh edition of MS-4 continues to offer chapters on types of asphalts, mix designs, mixing facilities, paving, compaction, surface treatments, recycling technologies, maintenance methods, rehabilitation, and structural design. But now, this all-in-one desk reference for asphalt highlights the latest technology.

New topics covered include the folowing:

- Superpave asphalt binder
- Superpave mix design
- Stone matrix asphalt
- Open-graded friction courses
- Quality control and acceptance
- Pavement management
- Rehabilitation of concrete pavements with HMA

New enhanced chapters include these topics:

- Petroleum asphalt
  - Asphalt modification and additives
  - PG grades
  - Safety guidelines
- Mineral aggregates
  - RAP
  - Mixture additives
- Thickness design
  - Mechanistic-empirical methods
  - Perpetual pavements
  - Comparison of design methodologies

And you will notice a new look for this new edition of MS-4. The seventh edition features an updated design and two-color layout, new graphics and photos, state-of-the-practice illustrations and charts, and a new topical/keyword index. Each chapter also includes a detailed reference list for more information when needed.

Acknowledgments

When the Asphalt Institute (AI) decided to give this flagship publication its first major overhaul in almost 20 years, we knew from the start that this would be a team-based project. The expertise required to accurately address such a broad range of topics simply necessitated this approach. We also knew that the sheer scale of the task invited the participation of others.

Our initial steps were to review the 1989 version of *The Asphalt Handbook* and to ask ourselves what the new handbook needed to include. Dr. Bob Boyer, P.E., former Asphalt Institute senior field engineer and now a consultant in Florida, led us through a yearlong process to develop a comprehensive and detailed outline for the manual, complete with references for each chapter. We are indebted to Boyer for his dedication to this fundamentally important stage of our project.

As we sought to build our writing team, we explored various alternatives with individual authors, partnerships, and companies, all of which added value to our process. We are indebted to them for their insights, advice, and remarkable excitement for what we were undertaking.

Selecting AMEC Earth & Environmental Inc., with Douglas Hanson as our content editor, proved to be the right choice for this project. AMEC's group of highly qualified experts with unique experience and knowledge combined with the Asphalt Institute's team to precisely deliver a new manual. We especially express our thanks to the following chapter authors for their expertise and dedication to this project:

- Douglas Hanson—Chapter 1, "Introduction to Asphalt"
- R. Michael Anderson—Chapter 2, "Petroleum Asphalt"
- Ken Wylie—Chapter 3, "Mineral Aggregates and Mixture Additives"
- Douglas Hanson—Chapter 4, "Hot Mix Asphalt (HMA) Design"
- Dr. Larry Hansen—Chapter 5, "Soils and Drainage for Asphalt Pavements"
- John Duval and Bob Humer—Chapter 6, "Thickness Design"
- T. J. Young—Chapter 7, "Manufacture of Hot Mix Asphalt"
- Thomas Skinner—Chapter 8, "Hot Mix Asphalt Paving"
- James Warren—Chapter 9, "Hot Mix Asphalt Compaction"
- Joseph A. Phillips—Chapter 10, "Quality Control and Acceptance of Hot Mix Asphalt"
- Kathryn Zimmerman—Chapter 11, "Pavement Management"
- Robert Campbell—Chapter 12, "Asphalt in Pavement Maintenance"
- Douglas Hanson—Chapter 13, "Rehabilitation Methods for Asphalt Pavements"

- Douglas Hanson—Chapter 14, "Rehabilitation of Concrete Pavements with HMA"
- Douglas Hanson and Robert Campbell—Chapter 15, "Special Uses of Asphalt and HMA"

Serving as the Asphalt Institute's project manager for the duration of this effort was Dr. Mark Buncher, P.E., our director of engineering. He developed the proposal requests, coordinated endlessly with Mr. Hanson on a myriad of technical details, and directed the execution of the peer review process. Buncher and Hanson approved the final products representing each chapter.

Providing the technical reviews for each chapter were the Asphalt Institute regional engineers: Mark Blow, P.E.; Sandy Brown, P.E.; John Duval, P.E.; Gary Fitts, P.E.; Bob Humer, P.E.; Mike Huner, P.E.; Wayne Jones, P.E.; and Carlos Rosenberger, P.E. These chapter reviews were critical to ensuring consistent and high-quality content throughout the manual.

Other key contributors on the Asphalt Institute staff included Dwight Walker, P.E., who managed editorial production and graphics and maintained the continuity of style and editorial consistency in all phases of the development process; Mike Sonnenberg, who managed the volume of electronic files and located countless images; Becky Boston, who kept track of the production schedule and coordinated printing for the new manual; Jacqueline Bartek, who researched photos and permissions; and Karen Embs, who helped proofread the manual. Brian Clark, Asphalt Institute's director of marketing and membership, oversaw the interface between AI's editorial team and our publishing partner, settingPace of Cincinnati, Ohio. Clark also developed the marketing strategy for the new manual and saw its implementation through.

The settingPace team proved invaluable as they adjusted to frequent schedule and process changes, turning our writing efforts and collection of images into professional, appealing pages. They were as much an essential part of this team as any other single component.

We are also indebted to the efforts of the Asphalt Institute Technical Advisory Committee and Marketing Committee for their contributions and advice along the way.

It takes a team and this team delivered! Thank you all for your efforts to publish the seventh edition of *The Asphalt Handbook*.

Peter T. Grass, P.E.
President
Asphalt Institute Inc.

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